

Thermal and Performance Analysis of Back-side Power Delivery Network beyond 3nm Technology Node

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Abstract — In this work, the Back-side Power Delivery Network (BSPDN) based on the NSFET and FINFET standard cell arrays are constructed for the thermal analysis in comparison of front-side power delivery networks (FSPDN). The performance metrics reflecting the IR drop and hotspot temperature are proposed to evaluate the characteristics of BSPDN. The impacts of different activity factors, inter-connecting metals and structure ratios are investigated. The results indicate that BSPDN faces the severer thermal challenges than FSPDN after the substrate thinning. And the design guidance of BSPDN is given for improving its performance at advanced node.

I. INTRODUCTION

With the continuous development of the advanced technology, the back-side power delivery network (BSPDN) [1-3] is implemented by using buried power rails (BPRs) [4-5] and TSV for improving IC integration. However, the narrow metal pitch and transistor dimensions exacerbate the thermal effects, causing the hotspot temperature increased. To clarify the thermal effects of BSPDN and balance the impacts of IR drop and temperature, in this work, we investigate the temperature distribution of NSFET and FINFET based standard cell array with BSPDN and FSPDN. The IR drop and hotspots are further discussed with different activity factors, metal materials and structure ratios.

II. SIMULATION STRUCTURE AND METHOD

Fig.1 (a) and Fig.1 (b) show the simulated structures of BSPDN and FSPDN. The standard cells based on the FINFET and NSFET structure with BPR at 3nm node are shown in **Fig.1 (c) and Fig.1 (d)**. The typical structure parameters and material thermal conductivities are listed in Table I and Table II respectively. The thermal simulation is achieved by FEM method with a 300K heat sink at the top and bottom of the frontside and backside interconnects. The initial heat powers are input based on the electrical properties of devices [6]. **Fig.2** shows the XY cross-sectional temperature distribution at fin/nanosheet regions with BSPDN and FSPDN (10% activity factor), which indicates that the overall temperature in NSFET is significantly higher than FINFET.

III. RESULT AND DISCUSSIONS

Thermal comparison of BSPDN and FSPDN: **Fig.3** shows the heat flux distribution of different PDN designs.

The backside heat flux contribution of BSPDN is significantly increased compared to FSPDN due to the worse heat conduction of backside interconnects, which should be paid more attention in thermal management. The hotspot temperature of BSPDN is lower than FSPDN with the large substrate thickness while the thermal effects of BSPDN becomes severer than FSPDN after the substrate thinning, as shown in **Fig.4**. From **Fig.5**, with the increasing activity factor, NSFET with BSPDN after substrate thinning faces the more serious thermal challenges than FINFET.

Performance metrics of BSPDN: **Fig.6** shows the impacts of VBPR and BPR with different metals on IR Drop. VBPR plays a decisive role in IR drop because its narrow cross-sectional dimension results in a large resistance [7]. The results show that VBPR and BPR with Ru is the optimal material with the lowest IR drop. The impact of different metals of BSPDN on hotspot temperature is shown in **Fig.7**. It can be seen that although Ru has the higher temperature, the performance metrics defining the product of ΔT and ΔV is significantly better than W, as illustrated in **Fig.8**. Due to the bad IR drop of W, the discussion in **Fig.9** about the impacts of varying ratios of Tsub and backside dielectric thickness (Tdielectric) mainly focus on the other two metal combinations. The hotspot using Ru is reduced with the increased ratio while the trend is opposite in the combination of Ru+W+Cu. **Fig.10** shows the normalized $\Delta T \cdot \Delta V$, which indicates that the performance of BSPDN with the combination of three metals is superior to Ru when its Tsub smaller than 3.5 μ m.

VI. CONCLUSION

Thermal effects of BSPDN and FSPDN based on the NSFET and FINFET standard cell arrays are compared in this work. The performance metrics reflecting the IR drop and temperature tradeoffs are further discussed. The heat flux distribution indicates that BSPDN needs more attention to thermal management on the backside. Furthermore, the impact of different metals and Tsub/Tdielectric ratios on both temperature and IR drop are given to provide the design guidance for BSPDN.

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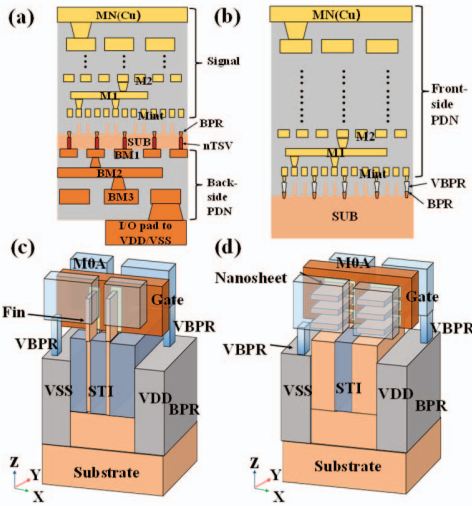


Fig.1 Schematic of (a) BSPDN with BPR, (b) FSPDN with BPR. The standard cells based on (c) FINFET and (d) NSFET.

	FIN	NSH	Layers	Pitch
channel H [nm]	50	5	BPR	105nm
channel W [nm]	5	24	M0	21nm
fin/nanosheet number	1	3	MINT	21nm
gate L [nm]	14		M1-M2	27nm
CPP [nm]	45		M3	33nm
μ TSV Diameter/Height [nm]	60/500		M4-M5	48nm
BPR H [nm]	130		μ TSV	500nm
BPR W [nm]	33		BM1	500nm
TiN liner [nm]	5		BM2	2 μ m
SiO ₂ /HfO ₂ [nm]	0.6/1		BM3	2 μ m

Table I The structure parameters and PDN design in this work for thermal simulation.

Thermal Conductivity	Value W/(K·m)
Fin/Nanosheet	[2 20 20] [20 20 2]
Source/Drain	36
Gate Oxide (SiO ₂ /HfO ₂)	0.4/0.5
Gate Pad	200
Silicon Substrate	130
Dielectric (SiO ₂)	2.5
M1-M5 (Cu)	380
BM1-BM3 (Cu)	380
VBPR (Ru)	117
μ TSV (Cu)	380
BPR (W)	178
TiN	4

Table II The material thermal conductivities used in this work for thermal simulation.

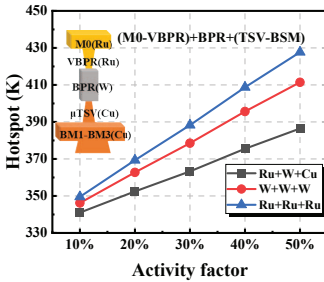


Fig.7 Hotspot temperature for different metal combinations of BSPDN with the increasing activity factor.

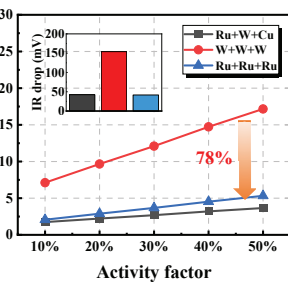


Fig.8 The performance metrics of $\Delta T \cdot \Delta V$ with different metal combinations.

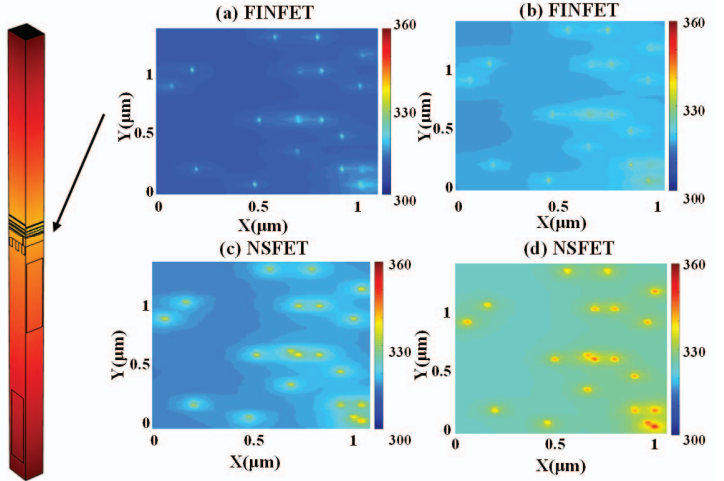


Fig.2 XY Cross-section of temperature distribution. FINFET: (a) BSPDN, (b) FSPDN. NSFET: (c) BSPDN, (d) FSPDN.

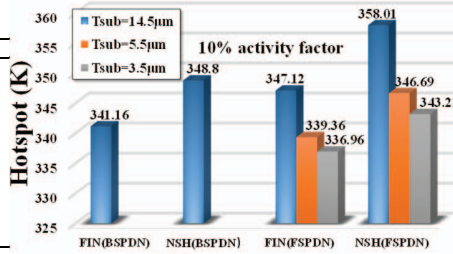


Fig.4 Hotspot temperature for BSPDN and FSPDN with substrate thinning.

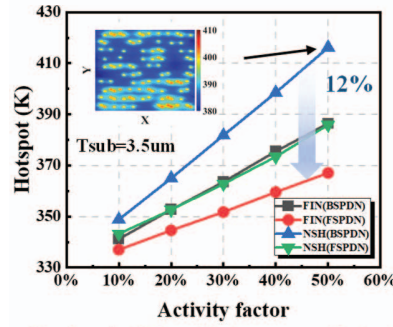


Fig.5 Activity factor vs Hotspot temperature for BSPDN and FSPDN based on FINFET and NSFET structure.

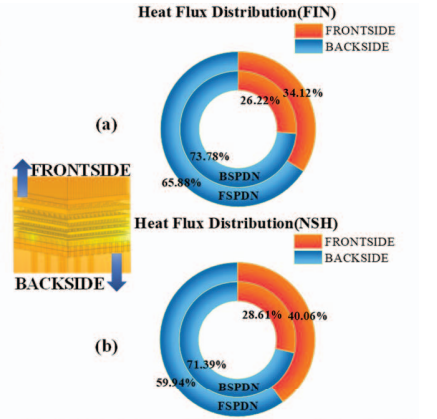


Fig.3 Heat flux distribution of backside and frontside interconnects with different PDN design. (a) FINFET, (b) NSFET.

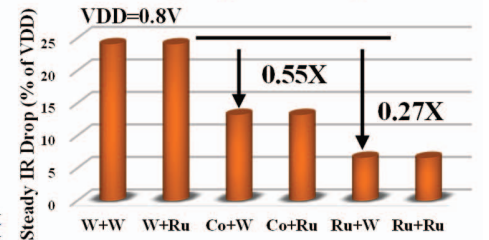


Fig.6 IR Drop of VBPR and BPR with different metals.

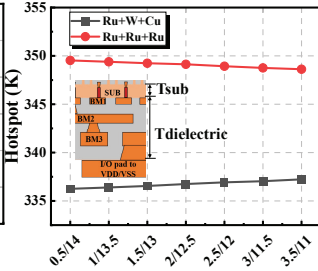


Fig.9 Impacts of varying ratios of T_{sub} and $T_{dielectric}$ on hotspot temperature.

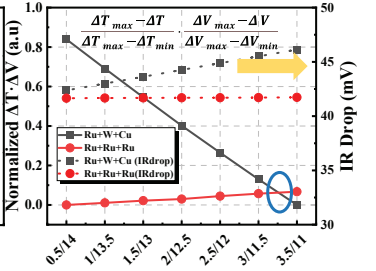


Fig.10 The normalized $\Delta T \cdot \Delta V$ with different ratio of T_{sub} and $T_{dielectric}$.